

shaft is made of steel having ultimate tensile stress of 750 Mpa and ultimate shear stress of 550 Mpa. If the factor of safety is 5.5 Find the diameter of the shaft. Assume any missing data .

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x30=30)

Q.26 Screw of vice has single start square thread, 22 mm diameter and 5 mm pitch. The diameters of collars are 55mm and 45 mm. Friction coefficients is 0.15 for threads and 0.17 for collar. Machinist can apply a force of 125N at the end of handle. The clamping force to be developed between jaw is 3000 N. Assuming uniform wear for collar find the length of handle required.

Q.27 A cam is to give the following motions to a knife edge follower.

- a) Outstroke during 60° of cam rotation.
- b) Dwell for the next 30° of cam rotation.
- c) Return stroke during next 60° of cam rotation.
- d) Dwell for the remaining 210° of cam rotation.

The stroke of the follower is 50 mm & the min, radius for the cam is 55 mm. The follower moves with uniform velocity during both the outstroke & return stroke. Draw the profile of the cam when the axis of the follower passes through the axis of the camshaft.

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4th Sem / Mech. Engg. (MSIL) Subject:- Machine Design and Drawing

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which of the following is not the method to reduce stress concentration?

- a) Usage of additional notches and holes
- b) Usage of fillet radius
- c) Usage of additional discontinuities
- d) Usage of sharp corners

Q.2 If a fastener is threaded into a tapped hole, then the fastener is likely to be called as

- a) Screw
- b) Bolt
- c) Washer
- d) Screw or bolt

Q.3 Resistance of a material against any external force is termed as _____

- a) Stiffness
- b) Malleability
- c) Strength
- d) Hardness

Q.4 Which of the following joins two rotating shafts to each other?

- a) Key
- b) Coupling
- c) Gear
- d) Belt drive

- Q.5 Which of the following is not a type of transmission shaft?
- a) crankshaft b) Countershaft
c) Transmission shaft d) Line shaft
- Q.6 Which of the following shaft are the integrals or equally spaced lines made on a shaft?
- a) Stepped shaft b) Spline shaft
c) Cam shaft d) Crankshaft
- Q.7 Which of the following does not form the important part of the screw jack?
- a) Frame b) Nut
c) Cup d) Coupling
- Q.8 Which of the following is a dominance of power screw?
- a) Large load carrying capacity
b) High efficiency
c) Wear of screw is reduced
d) Mechanical advantage is lowered
- Q.9 Which cam is also known as radial cam?
- a) Disc cam b) Cylindrical cam
c) Face cam d) Snail drop cam
- Q.10 Which of the following can be used for power transmission in intersecting shafts.
- a) Spur gear b) helical Gear
c) Bevel Gear d) None of the listed

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SECTION-B

Note: Very Short answer type questions. Attempt any ten questions out of twelve questions. (10x2=20)

- Q.11 Define Modulus of elasticity.
Q.12 Define Shear force.
Q.13 Define principal stresses.
Q.14 Write the angle between the principal planes.
Q.15 Give the formula by which angle of twist of a circular shaft is calculated.
Q.16 Describe bending.
Q.17 Describe feather key.
Q.18 Write the functions of power screw.
Q.19 What are different parts of a screw jack.
Q.20 Define pressure angle of a gear.
Q.21 Define simple harmonic motion.
Q.22 Give the relation between total depth, addendum and dedendum in a gear.

SECTION-C

Note: Short answer type questions. Attempt any two questions out of three questions. (2x20=40)

- Q.23 i) Define stress concentration and give the various methods to reduce it. (10)
ii) What is sunk key? Explain its different types. (10)
- Q.24 Draw four involute teeth of a gear 30 teeth of 10 mm module and 25° pressure angle.
- Q.25 A solid circular shaft is subjected to a bending moment of 3025N-m and torque of 10500 N-m. The

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